Worksheet 6.4 — Solving Trigonometric Equations

Grade 12 Advanced Functions (MHF4U) · Unit 6: Trigonometric Identities and Equations

The Big Ideas

Find all solutions on $[0, 2\pi)$: isolate the ratio, find the related angle, then use CAST for each quadrant.

- Isolate the ratio; find the related angle from a special value.
- Use CAST to place every solution.
- Quadratics: factor first; mixed: use an identity.

Core Concepts

Finding the angles

Solving a trig equation means finding every angle in the given interval that satisfies it, using reference angles and the unit circle.

Use identities first

Reduce the equation to a single trig function — often with a double angle or Pythagorean identity — before solving.

All solutions

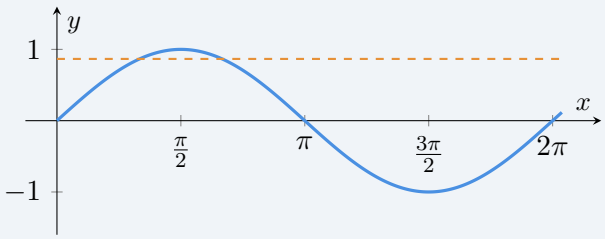
Because trig functions repeat, list every solution in the interval, and add the period for the general solution when asked.

Worked Examples

Example 1 — Basic sine

Solve $\sin\theta = \frac{\sqrt{3}}{2}$ on $[0, 2\pi)$.

Solution. Related angle $\frac{\pi}{3}$; sine + in QI, QII: $\theta = \frac{\pi}{3}, \frac{2\pi}{3}$. The wave reaches $\frac{\sqrt{3}}{2} \approx 0.87$ twice:



Example 2 — Cosine one

Solve $\cos\theta = 1$ on $[0, 2\pi)$.

Solution. $\theta = 0$ (the value 2π is excluded).

Example 3 — Isolate first

Solve $2\cos\theta + \sqrt{2} = 0$ on $[0, 2\pi)$.

Solution. $\cos\theta = -\frac{\sqrt{2}}{2}$; related $\frac{\pi}{4}$, − in QII, QIII: $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$.

Example 4 — Tangent

Solve $\tan\theta = \sqrt{3}$ on $[0, 2\pi)$.

Solution. Related $\frac{\pi}{3}$, + in QI, QIII: $\theta = \frac{\pi}{3}, \frac{4\pi}{3}$.

Example 5 — Quadratic type

Solve $4\cos^2\theta - 3 = 0$ on $[0, 2\pi)$.

Solution. $\cos\theta = \pm\frac{\sqrt{3}}{2}$; all four quadrants: $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

Example 6 — Basic cosine

Solve $\cos\theta = -\frac{1}{2}$ on $[0, 2\pi)$.

Solution. $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

Example 7 — Basic tangent

Solve $\tan\theta = 0$ on $[0, 2\pi)$.

Solution. $\theta = 0, \pi$.

Example 8 — Isolate first

Solve $\sqrt{2}\sin\theta - 1 = 0$ on $[0, 2\pi)$.

Solution. $\sin\theta = \frac{\sqrt{2}}{2}$; QI, QII: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}$.

Example 9 — Factor

Solve $(2\sin\theta - 1)(\sin\theta + 1) = 0$ on $[0, 2\pi)$.

Solution. $\sin\theta = \frac{1}{2} \Rightarrow \frac{\pi}{6}, \frac{5\pi}{6}$; $\sin\theta = -1 \Rightarrow \frac{3\pi}{2}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ on $[0, 2\pi)$.

Question 2

Solve $\cos\theta = \frac{\sqrt{2}}{2}$ on $[0, 2\pi)$.

Question 3

Solve $\sqrt{3}\tan\theta + 1 = 0$ on $[0, 2\pi)$.

Question 4

Solve $\tan\theta = -\sqrt{3}$ on $[0, 2\pi)$.

Question 5

Solve $\cos\theta(2\cos\theta + 1) = 0$ on $[0, 2\pi)$.

Question 6

Solve $\sin\theta = -1$ on $[0, 2\pi)$.

Question 7

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ on $[0, 2\pi)$.

Question 8

Solve $2\sin\theta + \sqrt{2} = 0$ on $[0, 2\pi)$.

Question 9

Solve $4\sin^2\theta - 1 = 0$ on $[0, 2\pi)$.

Question 10

Solve $\cos\theta = -1$ on $[0, 2\pi)$.

Question 11

Solve $\tan^2\theta = 1$ on $[0, 2\pi)$.

Question 12

Solve $\sin\theta = 1$ on $[0, 2\pi)$.

Question 13 — Challenge

Solve $2\cos^2\theta + \cos\theta - 1 = 0$ on $[0, 2\pi)$ by factoring.

Question Recap & Answer Key

Question Recap

1. Solve $\cos\theta = \frac{\sqrt{3}}{2}$ on $[0, 2\pi)$.
2. Solve $\cos\theta = \frac{\sqrt{2}}{2}$ on $[0, 2\pi)$.
3. Solve $\sqrt{3}\tan\theta + 1 = 0$ on $[0, 2\pi)$.
4. Solve $\tan\theta = -\sqrt{3}$ on $[0, 2\pi)$.
5. Solve $\cos\theta(2\cos\theta + 1) = 0$ on $[0, 2\pi)$.
6. Solve $\sin\theta = -1$ on $[0, 2\pi)$.
7. Solve $\tan\theta = \frac{1}{\sqrt{3}}$ on $[0, 2\pi)$.
8. Solve $2\sin\theta + \sqrt{2} = 0$ on $[0, 2\pi)$.
9. Solve $4\sin^2\theta - 1 = 0$ on $[0, 2\pi)$.
10. Solve $\cos\theta = -1$ on $[0, 2\pi)$.
11. Solve $\tan^2\theta = 1$ on $[0, 2\pi)$.
12. Solve $\sin\theta = 1$ on $[0, 2\pi)$.
13. Solve $2\cos^2\theta + \cos\theta - 1 = 0$ on $[0, 2\pi)$ by factoring.

✓ Answer Key

1. $\frac{\pi}{6}, \frac{11\pi}{6}$
2. $\frac{\pi}{4}, \frac{7\pi}{4}$
3. $\frac{5\pi}{6}, \frac{11\pi}{6}$
4. $\frac{2\pi}{3}, \frac{5\pi}{3}$
5. $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}$
6. $\frac{3\pi}{2}$
7. $\frac{\pi}{6}, \frac{7\pi}{6}$
8. $\frac{5\pi}{4}, \frac{7\pi}{4}$
9. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$
10. π
11. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
12. $\frac{\pi}{2}$
13. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$